



THE UNITED REPUBLIC OF TANZANIA
THE PRIME MINISTERS
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MBEYA CITY COUNCIL
MBEYA CITY JOINT EXAMINATION
FORM THREE PHYSICS



TIME: 3:00 HOURS

MARCH, 2026

INSTRUCTION

GEPAM science hub

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1. This exam consists of sections A, B and C
2. Answer all questions in section A and B, only two questions in section C
3. Mathematical table, graph papers and scientific calculator are allowed
4. Where necessary the following constants may be used:
 - (i) Acceleration due to gravity, $g = 10 \text{ m/s}^2$
 - (ii) Density of water = 1000 Kg/m^3 or 1.000 g/cm^3

SECTION "A" (16 MARKS).

1. Choose the best answer from the choices given by writing the letter of the correct answer
 - (i) The point where the Force of gravity can be considered to act is called:

A. Centre of gravity.	B. Centre of mass
C. Centre of weight.	D. Centre of body
 - (ii). If you are 180cm away from a plane mirror, what will be the position of your image?

A. 180 cm in front.	B. 360cm behind
C. 360cm in front.	D. 180 cm behind
 - (iii) Total internal reflection occurs when the following conditions are met EXCEPT
 - A. Angle of incidence is greater than the critical angle
 - B. The light is travelling from glass to air
 - C. The light is travelling from water to glass
 - D. The light is traveling from glass to air and vice versa

(iv) Which of the following is NOT true about the thin lenses?

- A. Convex lens is also called converging lens
- B. Concave lens is also referred to as diverging lens
- C. Convex lens can be biconvex, plano-convex or converging meniscus
- D. Most lenses are made of metals

(v). When a body floats on liquid:

- A. Weight is greater than the upthrust.
- B. Weight is equal to the upthrust
- C. Weight is less than the upthrust.
- C. Volume is greater than the upthrust

(vi) A force of 3.5 N acts horizontally on an object. Another force of 5N acts on the same Object at an angle of 45° to the horizontal. Which of the following is the resultant force?

- A. 8.2 N.
- B. 8.5 N.
- C. 6.1 N.
- D. 7.9 N

(vii). Khamis is poured liquid L into a measuring cylinder and noticed that concave meniscus
Which of the following is most unlikely to be liquid L?

- A. Ethanol.
- B. Water
- C. Soft drinks.
- D. Mercury

(viii). Magnet are often fitted on the doors of freezers and some cupboards. Why

- A. Because magnets used to keep away heat
- B. Because magnets used to keep inside's environment warm
- C. Because magnets used to keep the doors tightly closed
- D. Because magnets used to keep iron away

(ix). A boy weighing 4.5kg climbs up a stair case to a height of 7 m in 5 seconds. Calculate the amount of work done by the boy:

- A. 70J.
- B. 105 J.
- C. 450 J.
- D. 315J

(x) Refractive index can be determined using real-and-apparent method, where
refractive index equals

- A. Real depth/apparent depth
- B. apparent depth /real depth
- C. $\sin (I)/\sin (r)$.
- D. $\sin (r)/\sin (I)$

(2. Match the descriptions used in simple machines in **List A** with their corresponding terms from **List B** by writing the letter of the correct response beside the item number. (6marks)

List A	List B
(i) A device that multiplies effort force by two.	A. Inclined plane
(ii) A lever in which the effort is between the fulcrum and the load.	B. First class lever
(iii) The slope used to raise a heavy object with small effort.	C. Second class lever
(iv) A system that is assumed to have no energy loss.	D. Third class lever
(v) The ratio of distance moved by effort to distance moved by load.	E. Single movable pulley
(vi) A lever in which the fulcrum lies between the load and the effort.	F. Single fixed pulley
	G. Wheel and axle
	H. Efficiency = 100%
	I. Velocity ratio
	J. Mechanical advantage
	K. Friction
	L. Pulley

SECTION "B" (54 MARKS)

3. (a). It is more difficult to balance a nail on its tips than on its base. Explain why (03 marks).
- (b) Briefly explain why a drop of honey on glass sheets remains spherical while a drop of water spreads faster (03 marks).
- (c) A plane mirror is said to produce regular reflection. Explain why (03 marks).
4. (a) Briefly explain why concave mirror is used as shaving mirror or dentist's mirror (03 marks).
- (b) Explain why most vehicles have their engines directly over the drive wheels (03 marks).
- (c) Explain why hitting an inflated balloon with a hammer will not cause it to burst but

sticking it with a pin will (03 marks)

5. (a) What could be the effect on the resistance of a conductor if:
- i. its length was increased? (01 marks)
 - ii. its temperature was increased? (01 marks)
 - iii. its cross-sectional area was reduced? (01 marks)
- (b) Why is the knob on a door placed as far as possible from the hinges? (03 marks)
- (c) Give three reasons why mercury is preferred to be used as thermometric fluid (03 marks).
6. (a) Abu is a form two student. When he studies about simple machines, he gets an idea that they are there to simplify work, but he confuses their two concepts of "Mechanical Advantage (M.A)", and "Velocity Ratio (V.R)". Assist him to clarify these two concepts. (02 marks)
- (b) A machine having a velocity ratio of 5 requires 600 J of work to raise a load of 400 N. If the load moves a distance of 0.5 m, calculate
- i. Efficiency of the machine (03 marks)
 - ii. Mechanical Advantage (02 marks)
 - iii. Explain the type of machine is idea or non-ideal? (02 marks)
7. (a) Shortly explain why hot soup taste more delicious than cold soup (03 marks).
- (b) Briefly explain why houses are built or constructed with wide foundations (03 marks).
- (c) When we swim, we tend to spread out our legs and arms explain why (03 marks).
8. (a) Why car belts are made to stretch? (03 marks)
- (b) As a physicist brainstorm the reason why is it good to put load below seats than at the top of the bus (03 marks)
- (c) Which is easy; to open a nut by using a spanner with long handle or with short handle give reason (03 marks)

SECTION "C" (30 MARKS)

Answer only two(02) questions from this section

9. (a) What is the difference between monochromatic light and polychromatic light?
(03 marks).
- (b) show that to obtain an image with magnification of m , using a concave mirror with a focal length f , the object distance u , is given by $u = \frac{(m+1)f}{m}$ (06 marks)
- (c) A 4 cm tall light bulb is placed at a distance of 8.3 cm from the concave mirror having a focal length of 15.2 cm, determine the image distance and image size. (06 marks)
10. (a) During cold days we tend to rub our hands over each other to generate heat,
what force results to heat generation? give the meaning of that force (03 marks).
- (b) What are the ways that might be used to reduce the effect of the force above (list three ways) (03 marks).
- (c) Give the difference between fundamental and non-fundamental forces, with one example in each force (04 marks).
- (d) The rocket weighs 10000 N on earth moves to another planet X with the force of gravity of 1.56 N/kg. What will be the weight of the rocket on that planet? (05 marks)
11. (a). A boat is moving at 8.9 km/hr relative to the water in the river. Initially the boat at the bank aims straight to the opposite bank of the river which is 120.9m wide .if the speed of the water in the river is 2.9km/hr, how far downstream will the boat be when it reaches the opposite bank? (7.5 marks)
- (b) A nail drilled in the wall is being pulled out using a string ,the string form an angle of 30° with the normal.if the pulling force is 10 N, part of the force will tend to bend the nail while the other part will pull it out from wall. Determine the magnitude of these two components of the force (7.5 marks)

MBEYA CITY COUNCIL
F III MID TERM EXAMINATION
PHYSICS MARKING SCHEME

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1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	D	C	D	C	B	B	C	D	A

@ 1 mark

(10 marks)

2.

LIST A	i	ii	iii	iv	v	vi
LIST B	E	D	A	H	F	B

@ 1 mark

(6 marks)

3(a) Because its base has a wide base area and better stability, while the tip provides very very little support due to its small area. (03 marks)

- (b)
- Honey has strong intermolecular attraction than water
 - Honey has very high viscosity than water
 - Cohesion within honey is very strong, so it stays as rounded drop (03 marks)
 - For water, adhesion with the plate is often strong so water spreads out.

(c) Because its surface is smooth therefore, it produces regular reflection (03 marks)

4. (a) Because it produces a magnified, upright image. (03 marks)
Therefore, the image looks bigger.

(b) In order to reduce normal reaction so as to improve stability and transmission efficiency.
Note:
• It lowers the centre of gravity
• Improves vehicle balance (03 marks)
• Reduces the chances of skidding.

(c) Because the hammer has a large surface area and thus pressure becomes ^{low} high while a pin has a very small tip area and thus cause high pressure. (03 marks)

5. (a) (i) If its length increases, the resistance increases too.

(ii) If its temperature increases its resistance increases and thus lowers current. (03 marks)

(iii) If its cross sectional area was reduced, its resistance increases too.

5. (b) In order to increase the turning effect (moment of force), these enables to use small force to open the door with less effort. (03 marks)

(c) Because Mercury!

- It expands uniformly
- Does not wet glass
- It is clearly visible (03 marks)
- Good conductor of heat
- wide temperature range
 - Freezing point about -39°C
 - Boiling point about 360°C

6. (a) $MA = \frac{\text{Load}}{\text{Effort}}$ (01 1/2 marks)

$VR = \frac{\text{Effort distance moved}}{\text{Load distance moved}}$ (01 1/2 marks)

(b) Given, $VR = 5$ work input = 600 J
Load = 400 N, Load distance = 0.15 m

(c) Efficiency = $\frac{\text{work output}}{\text{work input}} \times 100\%$
 $= \frac{L \times L_d}{\text{work input}} \times 100\%$

$$6 \text{ (b) Efficiency} = \frac{400 \text{ N} \times 0.5 \text{ m}}{600 \text{ J}} \times 100$$

$$= \frac{200 \text{ J}}{600 \text{ J}} \times 100 \%$$

$$= \frac{2}{3} \times 100 \%$$

$$= \frac{200}{3} = 66.67 \%$$

$\therefore$ The efficiency of machine = 66.67%

$$(i) \text{ Efficiency} = \frac{MA}{VR} \times 100 \%$$

$$66.67 \% = \frac{MA}{5} \times 100 \%$$

$$\frac{66.67}{20} = \frac{20MA}{20} \quad (02 \text{ marks})$$

$$\therefore MA = 3.33$$

(ii) The Machine is not Ideal

Since its efficiency is

less than 100%.

(02 marks)

(a) Because hot soup has a lower surface tension than cold soup. Therefore, hot soup spreads over a large area of the tongue. (03 marks)

(b) Because pressure increases with depth. Hence, therefore are built in that way in order to withstand this pressure. (03 marks)

(c) • In order to increase upthrust (buoyancy) by covering large areas in order move water. (03 marks)
• To increase stability (to keep the body more stable)
• By third Newton's Law, pushing water backward → water pushes you forward.

(a) From, Newton's first Law of motion, seat belts are for safety against injury. Seat belts increase the time for impact, thus reducing the force on your body. (03 marks)
(Note, less force means less injury)

10

(c) Fundamental force Non Fundamental force

• Basic forces of nature

derived from fundamental force

(04 marks)

• Only four

Many types

• Act at atomic and cosmic levels

mostly seen everyday life

e.g. gravity

e.g. friction

(d) Given, $W_{\text{Earth}} = 10000 \text{ N}$
 $g_{\text{planet}} = 1.56 \text{ N/kg}$

$$W = mg$$

$$10000 = 10 \times m$$

$$\therefore m = 1000 \text{ kg}$$

(04 marks)

$$W_{\text{planet}} = m \times g_{\text{planet}}$$

$$10000 \div 10 = 1000 \times 1.56$$

$$= 1560 \text{ N}$$

The rocket weighs less on the planet because there is weaker.

8. (b) To Put Load below Seats:

- Lowers the centre of gravity (03 marks)
 - Increase the surface base area.
- Thus, increases its stability.

(c) It is easy to use a spanner with a long handle. The type of this move increases the turning effect (moment or torque). (03 marks)

Section C

9.

(a) Monochromatic Light	Polychromatic Light
• have one wavelength • One colour • Does not split into spectrum Example: Laser	- Many wavelengths (03 marks) - Many colours - Splits into spectrum (any two) e.g. Sunlight

10. (a) The force is called Friction (2 marks)

(b) Method of reducing friction:

- By using Lubricant e.g. oil, or water
- By ball bearings or rolling
- By using special materials

which have a low coefficient of friction e.g. Teflon. (0.5 marks)

- By polishing a surface to reduce roughness

- By using the materials with a high coefficient of friction e.g. tyres.

9 (b) Data given

Magnification = m

focal length (f) = f

object distance (u) = u

from the mirror formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad \text{--- (i)}$$

but magnification

$$M = \frac{v}{u}$$

$$v = mu \quad \text{--- (ii)}$$

Substitute equation (ii) into equation (i)

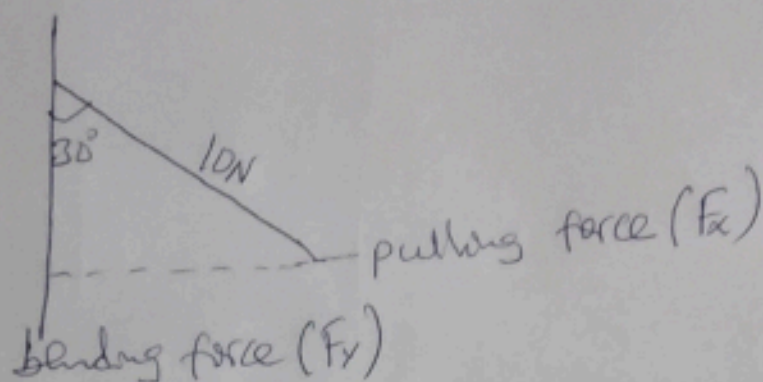
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{mu}$$

$$\frac{1}{f} = \frac{m+1}{mu}$$

$$\frac{mu}{m} = \frac{(m+1)f}{m}$$

$$u = \frac{(m+1)f}{m} \quad \text{hence shown}$$

11 (b)



from
 $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$

$$\sin \theta = \frac{F_x}{10N}$$

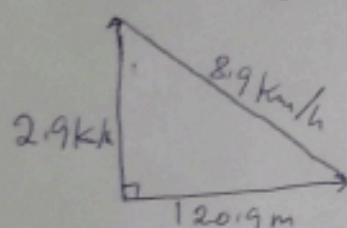
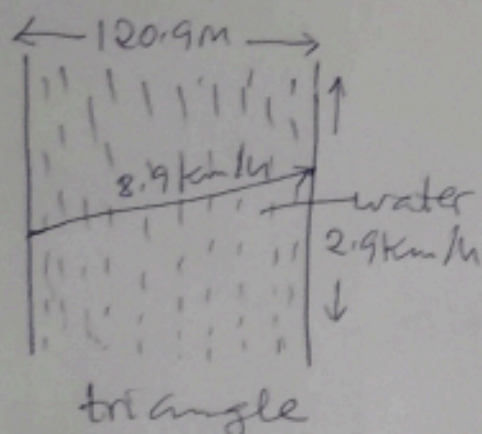
$$F_x = \sin 30 \times 10N$$
$$\text{pulling force} = \underline{5N}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{F_y}{10N}$$

$$F_y = 10N \times \cos 30$$
$$\text{bending force} = \underline{8.66N}$$

11 (a)



$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{0.1209 \text{ km}}{8.9 \text{ km/h}}$$

$$t = 0.0135 \text{ hrs} \approx 48.9 \text{ sec}$$

then

$$\text{distance downstream} = \text{speed of water} \times \text{time taken}$$

$$d = 2.9 \text{ km/h} \times 0.0135 \text{ hrs}$$

$$d = 0.03915 \text{ km}$$

$$\therefore \text{distance downstream} = \underline{39.15 \text{ m}}$$

9. (c) Data given

$$\text{Object size } (h_o) = 4 \text{ cm}$$

$$\text{Object distance } (u) = 8.3 \text{ cm}$$

$$\text{focal length } (f) = 15.2 \text{ cm}$$

$$\text{Image distance } (v) = ?$$

$$\text{Image size } (h_i) = ?$$

from the mirror formula

$$(i) \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{15.2} = \frac{1}{8.3} + \frac{1}{v}$$

$$\frac{1}{15.2} - \frac{1}{8.3} = \frac{1}{v}$$

$$\frac{8.3 - 15.2}{126.16} = \frac{1}{v}$$

$$\frac{-6.9}{126.16} = \frac{1}{v}$$

$$v = -18.28 \text{ cm}$$

$$\therefore \text{image distance} = -18.28 \text{ cm}$$

$$(ii) \text{ from } M = \frac{v}{u}$$

$$M = \frac{-18.28 \text{ cm}}{8.3 \text{ cm}}$$

$$M = 2.2$$

$$\text{but } M = \frac{h_i}{h_o}$$

$$2.2 = \frac{h_i}{4 \text{ cm}}$$

$$h_i = 8.8 \text{ cm}$$

$$\therefore \text{image size} = 8.8 \text{ cm}$$